

ADITYA

Backend Engineering · Full-Stack Development · Software Engineering

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EDUCATION

B.Tech Computer Science — Vellore Institute of Technology, Chennai

Aug 2023 – May 2027 | CGPA: 8.71 / 10

Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, Software Engineering, Cryptography

EXPERIENCE

Software Engineering Intern — Testing & Assurance | SAG, DRDO · Ministry of Defence, Delhi

May 2026 – Jul 2026

- Ran unit, integration, and static analysis (SAST) tests on classified defence-grade software; documented defects and remediation notes within a Secure SDLC workflow.
- Performed code review and software quality validation; applied secure coding practices and supported structured release planning across the development lifecycle.

Software Engineering Intern | Kirat Communication Pvt. Ltd., Haryana

Jun 2025 – Jul 2025

- Built a REST API backend in Flask for a multi-stage voice-AI pipeline (Whisper STT + Ollama LLM + Gemini API) on Raspberry Pi; cut end-to-end latency by 40% through request batching and audio buffer tuning.
- Took end-to-end ownership of MVP delivery — backend architecture, API design, implementation, testing, and deployment — shipping a working product in 6 weeks for a seed-stage startup.

Chairperson — Backend & Systems | Nexus Club, VIT Chennai (200+ members)

Apr 2025 – May 2026

- Led full-cycle development of club website and internal automation tooling (TypeScript, Next.js, Supabase) as sole developer — from requirements through production deployment.
- Ran Git/GitHub workshops for 200+ students; organised NexusForum with 1,800+ attendees, managing all technical infrastructure and event logistics end-to-end.

PROJECTS

CloudClip — Serverless Cross-Device Clipboard — AWS Lambda, API Gateway, Amplify, S3, JavaScript

- Built an event-driven serverless backend on AWS (Lambda + API Gateway + S3) for real-time cross-device clipboard sync; stateless REST endpoints with sub-100ms response time measured via Postman and zero-infrastructure auto-scaling.
- Designed for concurrent multi-device access without server provisioning — applying scalable cloud-native architecture principles to a real-world product use case.

Developer Portfolio — Full-Stack CMS [🔗 Live](#) — Next.js, TypeScript, Supabase (PostgreSQL), Tailwind CSS, Vercel

- Built and shipped a full-stack CMS as sole developer: Next.js frontend with SSR, Supabase PostgreSQL backend with row-level security (RLS), and a separate auth-protected admin dashboard enabling no-code content updates.
- Designed normalised PostgreSQL schema with role-based access control supporting 4 independent content modules (Blogs, Certifications, Projects, Life Journey) with real-time sync — zero redeployments for content changes.

RetailLens — E-Commerce Analytics Pipeline [🔗 GitHub](#) — Python, pandas, SQL, Streamlit, scikit-learn, matplotlib, seaborn

- Built an ETL pipeline in Python on a 500K+ row dataset — pandas cleaning, SQL aggregation, RFM segmentation, and cohort retention analysis; surfaced 34% repeat-buyer rate across 37 product categories.
- Shipped an interactive Streamlit dashboard with time-series visualisation and a logistic regression churn model at 81% accuracy; full EDA notebook on GitHub.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C, C++, Java, SQL, Bash | OOP, Data Structures & Algorithms

Backend & APIs: FastAPI, Flask, Node.js, RESTful API Design, Scalable Systems, Microservices, Containerisation (Docker)

Databases: PostgreSQL (schema design, RLS, query optimisation), Supabase, Firebase, MySQL

Cloud & DevOps: AWS (Lambda, API Gateway, Amplify, S3), Docker, Docker Compose, NGINX, CI/CD (GitHub Actions), Linux, SSH

Testing: Unit Testing, Integration Testing, SAST, Static Analysis, Code Review, Software Quality Assurance

Data & ML: pandas, NumPy, scikit-learn, matplotlib, seaborn, Streamlit, ETL Pipelines, EDA, TF-IDF, OpenCV

Frontend: React, Next.js, Tailwind CSS, SSR, Responsive Design

Tools: Git, GitHub, Postman, VS Code, Raspberry Pi, Agile/Scrum

CERTIFICATIONS & ACHIEVEMENTS

- LeetCode — 250+ problems solved; consistent across arrays, trees, graphs, and dynamic programming
- NVIDIA Deep Learning Institute — Fundamentals of Deep Learning (2025)
- Defy '25 Aptos Side Track Winner — blockchain smart-contract solution at national hackathon
- Chairperson, Nexus Club VIT Chennai — NexusForum flagship event with 1,800+ attendees; 200+ member community
- Open Source — 40+ public repos; Pull Shark & Pair Extraordinaire GitHub achievements